

BITS & PIECES

H8SCOOP REPRINT POLICY

I want to make known my policy regarding printing material found within H8SCOOP. I don't mind if other newsletters, HUG's or whoever wants to reprint material found in H8SCOOP if certain conditions are followed.

Anything printed from H8SCOOP should carry a notice from the issue it came from, along with the name, address and phone number of H8SCOOP and subscription information. This way, if someone sees something they like and want to subscribe they have the information available of where to subscribe and how much. That's all I ask. I feel this is fair as I'm not trying to hoard valuable information concerning the H8 and H89 systems, but would like credit for my efforts. I have noticed in the past, material has been distributed from H8SCOOP, especially on bulliten boards, without any mention of the source whatsoever.

ON TBT DIGITAL

Last month I reported in NEW STUFF things available from TBT digital, but I failed to mention the address and phone number. Ted Mieske, head of TBT is now back in Hurst, Texas. His phone number is (817) 282-3546. The address is 741 Oakwood Ave, Hurst, TX 76053.

ON THE SOFTWARE TRADER

Also an appology for last month's issue was the mention of The Software Trader neglecting the address. I have had lot of requests for it, so here it is: THE SOFTWARE TRADER, 423 Bedford Road, Schenectady, NY 12308. I'm very sorry for both of these address omissions.

ON QUIKMAIL DISK OFFER

For those who purchased the QUIKMAIL mailing label disk from me, if you are running the BH EXTENDED BASIC version, there is an error in line 4340. Instead of GOTO 4530 it should be 4350, many thanks to Ralph Munroe for pointing this out to me. In transcribing from MBASIC to BASIC, I entered this line wrong.

H8SCOOP SUBSCRIPTION PUSH

One of my goals for this year is to up the subscriptions of H8SCOOP, and I'm going to ask your help before I spend a lot of money in advertising. If any of you, my present readers can bring in new subscriptions, gather them and send them to me with payment, or have the individual mention

your name. I will extend the subscriptions for those who bring in new customers, the amount depending on the number of new subscribers you bring in. If you know of anybody, now is the time to get them to act. As the H8SCOOP circulation increases, everyone benefits. The more subscribers, the better service I can give. Advertising and classifieds reach more readers which helps all those who use the classifieds column or advertise under NEW STUFF, and the more interaction we can get for help and ideas.

CALL FOR HUG INTERACTION

I am also pushing this year for local HUG interaction. I would appreciate any feedback from local HUG's as to what your group is doing, and an exchange of ideas, which again, will help us all, and that's the name of the game. If you have a local HUG newsletter, send it to me. I want to know what's going on and keep in touch with as many HUG's as possible. Don't neglect your local HUG for my subscription push.

ON H89 HARD DISK DRIVES

Like I said, hard disk technology is here. Right now for about the same price as for dual 8" drives, you can get a formatted 5Meg hard disk drive from ACT, American Computer and Telecommunications Corp., 11301 Sunset Hills Rd., Reston, VA 22090 (703) 471-6288.

The 5 Megabyte Hard disk (Shugart Technology ST506) goes for \$1500, and the H89 controller card with CP/M Ver 2.2 goes for \$1250, for a grand total of \$2750. These, by the way, are assembled and tested units. The controller can control 2 of the ST506 micro Winchester hard disk drives simultaneously.

Hardware functions are accomplished with two separate boards. The larger board is located in the drive housing (typically, H-77 Heath variety). All functions of the controller are accomplished at this level with the exception of host buss interfacing, which is handled by the second card. This allows the user to move the ACT controller from one mainframe to another as needed by simply changing the system circuit card and user-controlling software.

I sent in to them for information and got back 19 pages (both sides) full of information, almost good enough for a users manual! I don't have the space to tell you any more here, if you want more info, contact ACT.

UPDATE ON DON LONG'S CLOCK BOARD

Don's CP-12 real time clock/calendar controller incorporates 5 Intel 8255A

Programmable Peripheral Interface (PPI) chips, each which has 3 parallel ports. These ports can be configured under software, for either input, output, or bidirectional data transfer. One PPI is dedicated for the Clock and board interrupt control, the other 4 being available for user I/O. It will be ideal for home control, energy conservation use and monitoring, burglar alarm applications and the like.

Don has the PC board layed out and should have a few prototypes by January for information. The boards should be available in quantity some time in February. 9319 Willowview, Houston, TX 77080.

CALL FOR HELP BY MEMBER

Ray Albrektson, 619 W. Dover, San Bernardino, CA 92407 was a victim of the arson fire in San Bernardino, California that was the big news on TV and the newspaper late in November. Ray's house happened to be in the way and burned down. Ray lost everything, computer, disk software, documentation, newsletters and everything else computerwise, not to mention the rest of the house and its contents.

Anyone with a heart that may want to help Ray get back on his feet, write him. He will need copies of software to replace that which was burned, documentation and anything else that goes with a computer. I have already sent him all back issues of H8SCOOP free of charge, and extend an invitation to Ray to let me know any software he lost. If I have it, a blank disk along with return postage will take care of it. Ray had an H89.

WAYNE GREEN BLASTS HEATH

I questioned the integrity of Sol Libes of BYTE, and now I wonder about Wayne Green, publisher of Microcomputing. I didn't have space to print this last month, but for those that missed it, I felt I would give you some highlights. It is from November, and on page 6. The gist of the editorial is selling computer systems with lousy software, especially to businesses. He has a point there, but listen to who's the blame.

"The S-100 bus was developing well, with the CP/M operating system making it look really good. Then we had the first really serious blow to the whole field- the Heath H8, with its own bus and own BASIC."

"There were music systems, talking systems, listening systems, control systems, plotters and modems for the S-100. The list was getting to be almost endless, and the star seemed bright for micros. All this went up in smoke when Heath, Commodore and Radio Shack hit the market with national

advertising, distribution in thousands of stores and virtually bare-bones systems. This threw two or so years of hard work and product development for most of the industry right out the window."

I dissagree the H8 caused any harm. The introduction of these systems had nothing to do with the S-100 or CP/M, if anything, make them better because of free enterprise competition! I was shopping for a computer right before Heath introduced theirs, and choose to go this way after evaluating what was available along with the prices, and influenced by the claims and promises of Heath.

For one who has Instant Software, which is an extreemely slow organization, and purchased a PRIME computer for all their mailing list work and other things, only to be totally let down, Wayne is the last one to be talking about computers. Enough said.

ON CBBS LISTINGS

Skip Prinsen, 3611 Merrimac Ave, San Diego, CA 92117, sent me a very complete updated CBBS MODEM set of phone numbers by Bill Blues in Santee. For those using your MODEM with CBBS interaction, contact Skip and ask him how you may obtain this listing. There are over 175 numbers on this list!

TIPS FOR THE H14 LINE PRINTER

Jim Czebiniak writes a method he used to allow his H14 to accept 15 lb. paper. He obtained a sheet of photo filter plastic, the kind they use for negative photo resist etching and cut off a small square (it's transparent red). This he said he slid between the paper sensor photo transistor and taped it in place. That's all there is too it. It allows a lighter bond paper to be used by fooling the sensor into thinking it's actually thicker.

Another tip for the H14. Jim says not to send more than 255 characters to the printer without a line feed or CR, otherwise the buffer will get overwritten from the beginning and this confuses the printer quite thoroughly. The result is that the print head goes wild and blows the .6A fuse on the circuit board.

ON QUESTIONNAIRE RETURNS

I am very pleased with the response to my questionnaires. Most of you that returned your questionnaires followed my philosophy and "told it like it is", and I appreciate that. All comments received are being evaluated and will receive my utmost attention in preparation of future H8SCOOP publications.

I appreciated the honest comments, both positive and negative on my writing and content. My goal is to mold H8SCOOP even more to comply with the wishes and requests of you, my readers as indicated in the questionnaire. On many issues there is about a 50/50 split, so these areas cannot be changed. One thing I've discovered long ago, you can't please all the people all the time, so you try to please the majority, and there will always be someone left out. As an example, about half wanted me to stop biking on Heath and throwing my own comments in, the other half said they were glad someone was "telling it like it is" and not afraid to print the truth, something other newsletters were shying away from. My policy stands, I will try not to needlessly cut someone or throw negative views where they don't belong, but when it is deserved, (my paying public deserves to know what's going on) I will tell what I know. I will try to avoid any over negativeness.

I'll get to points mentioned in the questionnaire and print answers as space permits. For this month the largest number of questions came in as to who the other newsletters are. BUSS, 325 Pennsylvania Ave, S.E., Washington D.C, 20003 is a good newsletter. While Charlie Floto, Editor of Buss was having problems keeping his newsletters on schedule (not near as bad as REMark), this seems to have been taken care of. The average ratings which came in, rated H8SCOOP highest, with BUSS right below. REMark was average rated very poor, most problems due to infrequency of publication and content.

BUSS was the first newsletter out after REMARK, and I, along with the rest of the Heath computer public, feel they got more of their money's worth from BUSS than REMARK. Subscription rates are \$18 per year, with most back issues available. BUSS, by the way, covers all Heath computers, ET-3400, H-11, H8, and H89.

Another newsletter many asked me about was PIP. I cannot speak for PIP since I do not subscribe, but the questionnaire rated PIP overall a 3-4 which is poor. For those interested, write PIP, POB 1126, Neptune, NJ 07753. That's all I know about this one.

The latest one on the market is M3 for Micro Media Magazine, 1316 Elmhurst, Garland Texas 75041. This is not really a newsletter, but a disk, so you can see right away, it is only for the H8/H17 or H89 systems. It can be ordered graphics (H19) or non graphics for other terminals. I believe the cost is something like \$12/issue, or \$55/year, it comes out once every 2 months. This one received a rating of 3-4. I have seen a few issues and feel you are paying too much for advertisements

and reports. The major negative feedback I have received on this one is you have to be in front of your computer and terminal to "read" it, and you most likely end up printing most of the text out anyway. There is software in each issue, usually clever utilities and the like, but I have not seen anything outstanding yet.

USERS COMMENTS ON DG PRODUCTS

"I have been running the DG-80 (Z80 CPU) along with two DG-32D (32K) memory boards for some time. So out of curiosity--more than anything else, I recently installed the DG-FP8 ROM and ADP4 4MHz adapter package. This includes new keypad identification labels--mine were poorly printed.

"The idea behind all this is--the ability to run CP/M at 000.000. But be forewarned, that to do so will cost you 8K of memory. I now have 48K as opposed to 56K. This is because unless you are running the DG-64D memory board a ROM disable port is required. It seems that on Boot-up, the FP8 ROM is transferred to high memory, so space in RAM must be left open to accomodate it.

"The DG-64D board has a built in ROM disable port so that one can use 56K of memory with HDOS and 64K with CP/M Ver 2.XX.

"DG Electronic Developments Company are supposed to have a ROM disable port (the DG-CMD1) for users of memory boards other than the DG-64D. But it is unavailable as of right now. AND THEY DON'T TELL YOU THIS IN THEIR ADDS.

"The new FP8 ROM does some neat things. The keypad entry and the corresponding front panel display can be in either octal or hex and they are instantly interchangeable. Nice. There is a built in 24 hour real time clock which resides in low memory and may be called up any time. The clock automatically updates the HDOS date at midnight.

"These are nice, but operating at 4MHz is really nice. Program execution time is cut in half. MBASIC programs load so fast from the disc that one can hardly hear the step motor click. Having the H8 operate at 4MHz really turns it into a professional machine.

"I am also running the H8 stretched, using Del Stanton's S T R E T C H-8. There is no problem running the stretched configuration at 4MHz."

This was sent in by William Velten, 12685 Parktrails Lane, Florissant, MO 63033. He also told me he made some improvements in the Stretch-8 and is very impressed with Del's product and service. William says if

anyone in the Saint Louis area would like to see a stretched H8 or need help with their kit, to contact him.

NEW STUFF

META TECHNOLOGIES, 26111 Brush Ave, Euclid, OH 44132 (Toll Free 800-321-3640) has available for \$24.95 a diskette storage box which will store 50 to 60 diskettes. It is a smokecolored acrylic unit which provides easy access through the use of index dividers and adjustable tabs. The lid provides dust-free protection and doubles as a carrying handle. The 5 1/4" version is \$24.95 and the 8" version is \$29.95.

More products and price updates from MINIT-MAN PRINTING, 211 E. Allegan St, Otsego, MI 49078. MF-001 VIDEO LAYOUT SHEETS used to layout graphics on the H19, H89 type terminals and computers. MF-002 GENERAL PURPOSE PROGRAMMING SHEETS used for basic and assembly language programming, back side carries info on Heath terminal escape sequences, plus area for user defined variables. MF-003 FLOWCHARTING WORKSHEET carries a 5X9 array of programming positions. Each position has 5 general programming symbols plus connecting lines on center and between each position, both vertical and horizontal allow easy flowcharting without the use of templates.

All forms are printed in fade-out blue on 60 pound white offset stock and padded in 50's, and standard 3 hole drilled for ring binders. The forms are \$2.25/pad, mix or match, with a minimum of 5 pads, which includes shipping.

Walt Bilofsky of Software Toolworks, 14478 Glorietta Dr., Sherman Oaks, CA 91423 has a RATFOR language processor and UTILITIES disk.

The RATFOR disk which requires the Microsoft FORTRAN sells for \$39.95. See past H8SCOOP's to find out what RATFOR is all about. The disk includes documentation, sample programs, I/O library, and full source code for the RATFOR translator written in RATFOR.

The UTILITY disk for \$24.95 contains a Master Catalog System to keep track of your disk files, by printing to printer, console, or file, all files which satisfy "wild card" conditions or with specified flags or creation date limits. Five utility programs are included. FIND selects lines in specified text files which contain a specified string or patterns. CHANGE replaces one string or pattern in a file with another. CMP does a byte by byte compare of two files. CHECK prints the CRC checksum of specified files, and FDUMP is a byte by byte file dump with up to 5

selectable formats.

These will run on 32K H8 or H89 under HDOS 1.5 to 2.0. Add \$2 per order for first class postage.

Microcomputing pulls through again, with a short 1 page article written by our own D.C. Shoemaker (just call him Major) on "PROGRAMMING HEATH'S USART", on page 216 of December's issue. This covers the H8-5 serial board which uses the INS8250.

L. D. BARRON, 1441-B N. Red Gum, Anaheim, CA 92806 has a disk offer for \$16.95 post-paid USA (foreign add \$2). It's called LOAD-N-GO. It modifies HDOS 1.6 (issue 50.05.00) to provide for automatic boot-up for any BAUD rate from 600 to 19200 user selectable, auto-mount of validly installed SY1: and SY2: disks, and run AUTO or AUTO.BAS on both cold AND warm boot. You can now mix machine-code, MBASIC, and BHB programs on the same disk and run them from a single menu or controlling program (samples provided). LOAD-N-GO is fully compatible with HDOS STAND-ALONE and SYSMOD.

It also includes a program to PATCH your disk directory for a 4 sector directory thus saving 14 sectors, and still leaving room for about 40 directory entries. The disk comes in a plastic packet similar to the way Walt Bilofsky ships his software. It includes mini-instructions on paper with the pack, and full instructions on the disk. Write L. D. Barron for more information or to place an order.

REQUESTS

Bengt Thorson M.D., S-51020 FRITSLA, SWEDEN would like to know if anybody has any information on the BSR controller for 220Volts, 50 Hz. If you have any info on this, write to Bengt in Sweden, He's a very sincere computer enthusiast, who would like support from the States.

A request from Ted Mieske, 741 Oakwood, Hurst, TX 76053 (817) 282-3546. Ted has the Heath Weather station and has it interfaced with his H8. The way it is set up now, TMI must be used, which terminates HDOS. Ted wants to use this with HDOS so he can use it with a MODEM phone answering bulliten board service he is working on. He wants to know if anyone has a way to interface this weather station to the computer under HDOS so he can get the weather information and still remain in HDOS, to make the data available to MODEM users who will be calling his bulletin service in the future. Any info, direct it to Ted.

From Bill Hall (see address in classifieds): Are there any H8 users in the Pittsburgh, PA area? Would you like to exchange HUG and other software?

Robert Mehaffey, 6001 Winchester Park Drive, New Orleans, LA 70128 has a request for the readers. He has a TI-810 line printer which counts 62 lines and skips to the top of the next page. He wants to know if there is a software command that will disable this function or some other fix. Any info, write to Bob.

Henry Beechhold, 13 Perry Drive, W. Trenton, NJ 08623 has the SWTP CT-82 CRT terminal, which he says is a 9" screen green phosphor, sharp as a photograph. He would like to know if anyone has configured H19 graphics stuff to run on the CT-82.

WHO'S WHO

Jim Bates (see Classifieds for address) runs an H9 connected directly to a PENNYWHISTLE Modem. He says all that must be connected up are: Signal IN, Signal OUT, and Ground. It works fine as a Dumb Terminal, comments Jim.

OFFERS

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested.

I make all software available on an as-is offer, in other words, no money back policy if not satisfied, mainly because it's too easy to copy disks. I try to get software out as fast as possible, and as thus, don't get a chance to carefully screen and test everything. If I did, all software would be delayed 3 to 4 months. I do the best I can, and sell with assuming no responsibility on the accuracy or runability of a program or disk. I don't think I've let anyone down in the past (not at my prices) and will continue to do my best not to.

TINY PASCAL	\$10
SEPT DISK--FILE DUMPS & UTILITY	\$ 7
NPS CP/M MICRO COBOL SET(2)	\$25
NPS CP/M MICRO COBOL SAMPLE TEST PGM	\$ 5
RATFOR TRANSLATOR	\$ 7
NOVEMBER DISK--PICTURES, LABEL, ETC	\$15
QUIKMAIL MAILING LIST PROGRAM	\$15
C SOFTWARE STUFF DISK	\$10
DECEMBER DISK--PICTURES & UTILITY	\$10

To answer a question many H89ers have written in, YES, these disks are all compatible with the H89 system. HDOS is HDOS, whether on the H8 with the H17 or the H89, the operating system is all the same.

A question on the NOVEMBER disk, DRAW.MB. You may be getting an out of memory message if you do not have at least 52K. When you run MBASIC, gobs of memory are consumed and it appears you need lot of memory to run the thing.

This month I am offering a special H19 graphics disk, JANUARY OFFER DISK I which contains quite a few of the stupid fascination programs by John Welch, 1306 Elm Dr., Schaumburg, IL 60194. It also contains other H19 graphics programs for automation, such as WORM, KALIDESCOPE, PINBALL.ABS (game), listing program and others. Many require MBASIC. This one goes for \$10.

TECHNICAL FORUM

MORE ON HDOS 1.6 SYSTEM

This month we will use a DUMP program to remove and change flags in the disk directory under HDOS version 1.6. Most of this information available by John Merrill.

If you look at the directory of the disk starting at track 22 (0-39) and sector 2 (0-9) you will observe the following format for each directory entry.

```

0          1          2
12345678 901 234 5 67890123
FILENAME EXT XXX F XXXXXXXXX

```

```

Block 1 to 8   is FILENAME
Block 9 to 11  is EXTENSION
Block 15       is FLAGS

```

If you look at the number in the flags block (15) in binary, you can determine which flag(s) is set, if any using the following chart. You can also use DUMP or a similar program to change these flags, including removing the Lock flag.

FLAGS:		
FLAG	HEX	BINARY
		SLWC
		11110000
NONE	00	00000000
C	10	00010000
W	20	00100000
L	40	01000000
LW	60	01100000
S	80	10000000
SW	A0	10100000
SL	C0	11000000
SLW	E0	11100000
SLWC	F0	11110000

What's the C flag? It stands for Contiguous. It tells HDOS the file is in order and does not need the linked list to tell it where to jump for the next file

part. You say you've never seen the C flag? It's hidden. Type CAT/JGL and you will. It is really of no concern to you, only to HDOS. Files with a C flag can be read faster since they are in order and there is no jumping around. This also means less disk activity. The C flag files are all system files, HDOSOVLO, HDOSOVLL, HDOS, RGT, AND GRT. Just remember if you remove the L flag to keep the C flag.

As long as we're on Flags and files, just for your information: GRT is the Group Reservation Table. It is associated with locating the parts to each file. RGT is the map which contains any bad sectors that the directory should not use. When you do an INIT you are asked for bad sectors, well, this is where they go. When the directory is accessed for write, HDOS first checks RGT to determine if there is an area that the directory should not use for storing files.

DISK SPIN HDOS DISK LOCATION

From L. D. Barron on the disk spin clock (See H8SCOOP September and October 1980). He writes that the byte can be found in the second sector of HDOS.SYS. With HDOS in normal position, and using a DUMP program (0-39 TRACK and 0-9 SECTOR), byte at track 1 sector 3 location 70H. The normal value is 3CH. Changing it to 0AH results in a spin time of about 5 seconds, while a change to 14H results in a 10 second spin time.

BOOT DISKS WITH TOO HIGH STEP RATE

Here's some neat info sent in by Ryck Zarick on how to boot up a disk that may be set to step too fast for your drives.

First boot up a disk that has SET.ABS on it and copy SET.ABS onto the disk that will not boot up because the step rate is set too fast.

Now SET HDOS STAND-ALONE and RESET SY0:. Remove the disk you copied SET.ABS from and insert your disk you cannot boot into SY0:. Next SET SY: STEP XX, where XX is whatever your drives step at. Now type BYE to get the disk dismounted and insert your first disk again and SET HDOS NOSTAND-ALONE.

You're back to the HDOS operating system, and your disk should now boot.

ON H19 DISK DRIVE MOUNTING

I have received some info on mounting an extra drive for the H17 in the H19 cabinet as mentioned in December 1980 H8SCOOP. First several readers informed me that the flyback transformer in some of the H19's will damage or inadvertently write bad data on the diskette when the drive is placed in close proximity to it. So beware. Some

shielding may be necessary. Since the H89 does it, it stands to reason the H19 can also do it. They both contain the flyback transformer in the same proximity to the disk drive location.

I was also warned that the power supply in the H17 may be marginal, even though Heath originally stated it could handle 3 drives. I understand their 3 drive mod places a fan in the H17 along with a hefty power supply diode bridge to beef up the power supply a little, and a larger filter capacitor.

Anyway, two individuals responded to my write-up, Ralph Monroe and William Velten. William has already done it and says it's working. He sent me the following instructions of what to do. He has the computer to his left, the H19 in the center and the H17 to the right of the H19, which places the 3 drives side by side, with SY0: in the H19.

To install the drive you will need from Priority One Electronics, 16723 Roscoe Blvd, Sepulveda, CA 91343 (300) 423-5633--10 feet color coded 34 conductor ribbon cable, P/N IDC-34CC; one socket connector for the controller card in the H8 (solderless) P/N IDC-34SKT; and three edgecard connectors for Siemens drives (solderless) 3 each P/N IDC-34CE.

Install the IDC-34CC on one end of the ribbon cable. Install a IDC-34SKT in the middle of the ten foot ribbon cable and the other two connectors approximately 6 inches apart on the other end.

In addition you will need about 5 feet of 16 guage 3 conductor flat or twisted cable to provide power to SY0: in the H19 from the H17 power supply. Then connect 12V, 5V and ground on the H17 power supply board.

The following parts can be ordered from Heath:

- One 206-1331 Mu-Metal Shield
- One 204-2420 Top Floppy Bracket
- One 204-2421 Bottom Floppy Bracket
- One 205-1804 Support Bracket
- One 75-124 Large paper insulator
- One 250-56 6-32 X 1/4" screw
- One 254-1 #6 lockwasher
- One 73-166 Edge Strip
- One 75-790 Insulator paper
- One 434-319 Connector housing
- Three 250-1264 6-32 X 3/8" hex head screws
- Three 75-108 Small Paper insulator
- Four 432-1002 Female connector pins

Obtain a 6" piece of 16 black wire to go from the spare ground in the power supply connector on the disk drive to the ground lug near the bottom of the large vertical capacitor on the H19.

William says once you've got all the parts together it should not be too difficult to figure where everything goes. Anyone needing help feel free to contact William Velten at 12685 Parktrails Ln, Florissant, MO 63033. His phone number is (314) 355-4072.

BOOK REVIEW

PASCAL

I am reviewing a book this month I feel very timely since Heath is about to release their PASCAL, and the MT PASCAL is available for the H8. CP/M PASCAL has been available for some time for the H8. So now that it's here, what is it?

PASCAL, by Paul M. Chirlian is an excellent book on the language in general. It was not written for the H8, but takes a novice through the entire language. If you cannot obtain it from your local book or computer store, write ISBS INC., 2130 Pacific Ave., Forest Grove, OR 97116. The price on it is \$9.95

The book emphasizes the structured nature of PASCAL. If you have not had any experience with structured languages such as C, the book gives enough of a background and examples for you to pick up on it.

Most every aspect of PASCAL is covered in this book; Arithmetic operations, I/O, Control Statements, Arrays, Characters and Strings, Subprograms, Files, Sets, RECORD and Variant, and Dynamic Data Structures.

The beginner as well as one desiring a cookbook will benefit from this text. I quote from the Preface since it sums up my feelings in better words than I could produce. "The subject of Pascal is discussed rather completely. Thus, when the beginner has gone through the text, he will be familiar with most of the aspects of PASCAL. It is hoped that the book will serve as a reference so that even advanced programmers can refer to it to refresh themselves on some details. To aid in this, several appendices indicate the salient points of PASCAL.

"The book is organized so that a beginner can start running programs almost immediately. Many varied exercises are included at the end of each chapter".

The book is filled with examples. It presents a point, or KEYWORD of the language, and then gives an example of how it is used in a program. There are many example programs throughout the book.

I feel the book is well written and if the user has read this book and has access to PASCAL, he will be well on the way to learning and efficiently using the language. While all the examples and KEYWORDS given in the book may not apply to your H8 version, enough will be learned including limitations of special functions not included with all computers, to get you on your feet. END REVIEW

Henry Beechhold writes in to recommend another book for Supersoft's Tiny Pascal (TRS-80 implementation), which is very similar to the Tiny Pascal Jim Teixeira decoded from Chung and Yuen's version just as Supersoft did. The book by David L. Heiserman is called PASCAL, and it's a Tab book. Henry said he got it as a come-on for joining Tab's Electronics Book Club. He says he is impressed with Tab books as opposed to Sams.

* PROGRAMMING GOODIES *

A short program sent in by John Welch, a Fascination program is printed here. John says a true stupid fascination program (SFP) should be short, quick to write, and stupidly fascinating. They usually do not ever end. The one presented here is SNOWSTORM. Available as my January offer is this, plus other SFP programs of John's.

SNOWSTORM is written in MBASIC for an H19 graphics terminal. An H89 will also work. I see no reason why it won't run in BH EXT basic if you remove the CLEAR statement in line 10, and remove all % symbols, which stand for INTEGER in MBASIC. Here it is.

```
10 PRINT CHR$(27); "E": CLEAR 1000:
   DIM M%(24,80), S%(20,2)
11 FOR X=1 TO 80: M%(23,X)=1: NEXT X
20 DEF FNR(X)=INT(RND(3)*X)+1
30 FOR X=1 TO 15: S%(X,1)=FNR(74): NEXT X
40 FOR Y=1 TO 22: FOR X=1 TO 15: IF S%(X,2)
   =1 THEN 110
45 D1=FNR(3)-2: IF S%(X,1)+D1<0 THEN D1=-D1
50 IF S%(X,1)+D1>74 THEN D1=-D1
60 IF M%(Y,S%(X,1)+D1)=1 THEN S%(X,0)=
   Y-1: S%(X,2)=1: M%(Y-1,S%(X,1))=1: GOTO 110
70 PRINT CHR$(27); "Y"; CHR$(31+Y-1); CHR$(
   31+S%(X,1)); " "
80 PRINT CHR$(27); "Y"; CHR$(31+Y); CHR$(
   31+S%(X,1)+D1); "*"
90 S%(X,1)=S%(X,1)+D1
100 S%(X,0)=Y
110 NEXT X
120 NEXT Y
130 FOR T=1 TO 15: M%(S%(T,0),S%(T,1))=1:
   S%(T,2)=0: NEXT T
140 GOTO 20
```

For those beg-nners, here's a fast one to print 70 dashes across the screen, ideal for a fast chart or heading underline, or whatever:

FOR X=1 TO 70:PRINT "-";:NEXT X:PRINT

Just a quicky, to obtain Pi to an accuracy of .0000085% use the following value of Pi=355/113.

Submitted by Alden G. Finley, 1429 Balmy Beach Drive, Apopka, FL 32703 is A DO-ALL, NEAR FOOL-PROOF, MBASIC LOADING PROGRAM. He writes "The combination of the two programs, M and MENU, provides the HDOS/MBASIC user with a near fool-proof, do-all, loading sequence. Both programs should be on your system disk which is mounted in SY0:. MENU5M.BAS must be renamed MENU.BAS.

"Upon pressing "M" and the CR, the SUBMIT or DOCOM sequence starts. The HDOS version number is listed and your pointer is LOADED. MBASIC is loaded and then, in turn, MBASIC loads and RUNs MENU.BAS.

"MENU.BAS first checks to see if there is a diskette mounted in SY1:. If not, you are asked to install a diskette in SY1:, or if you plan to use the diskette in SY0: alone, you can enter a CNTRL A and proceed. Assuming that you did want to use SY1:, the program first looks for a MENU.BAS on the SY1: diskette. This menu program must be different from the MENU.BAS in SY0: and is to be provided by you to execute programs on the SY1: diskette or what ever you choose for it to do. If your menu program is found, it is RUN and you go from there. If it is not found, then the program asks you to input the brief (no drive or extension required) filename of the program you wish to RUN. If you enter "R" as the filename, then SY0: will be RESET so that you can install a new diskette in that drive while in MBASIC. There are other features that will become apparent upon use. Ample error traps are included to assure a near fool-proof operation.

"It is left to the reader to modify the program for inclusion of SY2: or any other special actions desired. Your comments and suggestions are welcome.

```
10 ' MENU5M.BAS A. G. FINLEY 20-SEP-80
20 ' MBASIC PROGRAM CALLED BY SUBMIT
   PROGRAM "M"
30 ' MUST BE NAMED MENU.BAS AND BE ON
   SY0: DISK
100 ON ERROR GOTO 2000
250 F1$="SY1:MENU.BAS" 'INDIVIDUAL MENU
   PROGRAMS
251 F2$="SY1:DIRECT.SYS"
500 OPEN "I",1,F2$
510 CLOSE
520 GOTO 1000
580 PRINT: PRINT"INSTALL DISKETTE IN SY1:,
   OR PRESS <CNTRL-A>."
590 RESUME 600
600 RESET "SY1:"
```

```
1000 '
1010 RUN F1$
1020 PRINT
1030 PRINT "ENTER BRIEF FILENAME OF MBASIC
   PROGRAM DESIRED"
1031 PRINT "OR TYPE <R> TO RESET SY0: ";
   :LINE INPUT A$
1035 IF A$="R" THEN RESET"SY0:":NEW
1040 RUN A$
1050 RESUME 1060
1060 A$ = "SY1:"+A$:RUN A$
1070 GOTO 1030
2000 IF ERR=53 AND ERL=1010 THEN 2020
2001 IF ERR=54 AND ERL=1010 THEN 2020
2002 IF ERR=65 AND ERL=1010 THEN 2020
2003 IF ERR=65 AND ERL=500 THEN 580
2004 IF ERR=65 AND ERL=1040 THEN 1030
2005 IF ERR=53 AND ERL=1040 THEN 1050
2006 IF ERR=53 AND ERL=1060 THEN 2040
2010 ON ERROR GOTO 0
2020 PRINT:PRINT F1$; " CAN'T BE FOUND."
2030 RESUME NEXT
2040 RESUME 1020
```

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FOR SALE--H9, asking \$135, H16 16K memory board \$175, will sell both for \$275. Thomas Bessire, 5817 Boca Raton, Apt. J108, Ft. Worth, TX 76112. (817) 451-3912.

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8

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